

2010 HSC Notes from the Marking Centre — Chemistry

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Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Chemistry. It contains comments on candidate responses to the 2010 Higher School Certificate examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

This document should be read along with the relevant syllabus, the 2010 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Chemistry.

General comments

Teachers and candidates should be aware that examiners may ask questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course including the prescribed focus areas.

It is important to understand that the Preliminary course is assumed knowledge for the HSC course.

Candidates need to be aware that the marks allocated to the question and the answer space (where this is provided on the examination paper), are guides to the length of the required response. A longer response will not in itself lead to higher marks. Writing in excess of the space allocated may reduce the time available for answering other questions.

Candidates need to be familiar with the Board's [Glossary of Key Words](#) which contains some terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the key words from the glossary. Questions such as 'how?', 'why?' or 'to what extent?' may be asked or verbs may be used which are not included in the glossary, such as 'design', 'translate' or 'list'.

Teachers and candidates are reminded that mandatory skills content in Module 9.1 are examinable in both the Core and Option questions.

Candidates should use examination time to analyse the question, plan their responses carefully, and then work within that framework to produce clear, logical and concise responses. The response may include the use of dot points, diagrams and/or tables and planning the response will help to avoid internal contradictions. Holistic responses need to be logical, well constructed and relevant to the questions asked.

In better responses, candidates:

- set out all working for numerical questions and did not round off until the final answer
- thought carefully about the units to be used and the quantities to be substituted into formulae
- did not repeat the question as part of the response
- looked at the structure of the whole question and noted that in some questions the parts followed on from each other, eg responses in part (a) led to the required response in part (b)
- used appropriate equipment, eg pencils and a ruler to draw diagrams
- accurately transcribed and used values from the periodic table and data sheet for calculations
- included balanced chemical equations where appropriate with the correct states of all species shown
- were guided by the space provided on the paper as to the length of the response necessary.

In Section II, each Option question is divided into a number of parts. Candidates should clearly label each part of the question in their writing booklet, and follow the instructions to answer in a separate writing booklet. Candidates are strongly advised to answer the Option they have studied in class.

Section I – Core

Part B

Question 21

Better responses explained that the weak acid partially ionises and the strong acid fully ionises even though the concentration of the hydrogen ions is the same for both acids. In weaker responses, there was confusion between strength and concentration with some responses stating the stronger acid is the more concentrated.

Question 22

- b. Better responses explained that sulfuric acid increased the rate rather than the refluxing.
- c. Weaker responses identified propanoic acid but had drawn the structure for butanoic acid, not counting the carbon in the functional group as part of the name.

Question 23

- a. Stronger responses included a balanced chemical equation for the complete combustion of butan-1-ol. A common error in the weaker responses was not counting the oxygen atom in the butan-1-ol when balancing the equation.
- b. In better responses, candidates justified the choice of fuel from a table of results of a practical experience by including calculation/s with step-by-step working as well as reasons for selecting the fuel. They included loss of heat and incomplete combustion as reasons for the discrepancy between the theoretical and experimental value for butan-1-ol. They linked knowledge of heat of combustion to the practical experience. The better responses recognised that a calculation can form part of justification for the selection of the fuel.

Weaker responses did not show clear working for calculations and tried to match the answer for the calculation directly to the heat of combustion of the listed fuels.

Question 24

- a. In better responses, candidates included a balanced chemical equation for the hydrogenation of ethylene to produce ethane and stated that the hydrogenation reaction was an addition reaction. Weaker responses made the error of misreading the word 'hydrogenated' as 'hydrated' or 'halogenated' leading to incorrect equations. Weaker responses either included the correct equation or stated the correct type of reaction only. Many weak responses had errors in the balancing of the equation.
- b. In better responses, candidates stated that the colour change of the bromine water is from brown to colourless in the presence of ethylene, and if the ethylene has been fully converted to ethane, the bromine solution would remain brown. These responses also stated that if the ethylene is not fully converted, the bromine water would change from brown to colourless as ethylene is still present.

Question 25

Better responses were comprehensive, coherent and logical, and gave a clear description of biochemical oxygen demand (BOD) and dissolved oxygen (DO) quantitatively, relating them to each other with an explanation of why each needs to be monitored. Mid-range responses were less comprehensive in addressing all the parts of the question and only had an outline of the DO and BOD. In such responses, only simplistic links between BOD and DO were provided with an explanation for the need to monitor only one of them. Weaker responses communicated only simple ideas related to BOD and DO. Most of these responses showed a lack of understanding of BOD.

Question 26

The best responses clearly set out working, provided a balanced chemical equation and included correct states. Zinc was recognised as being in excess and nitric acid was described as a limiting reagent. Most responses contained accurately transcribed values from the periodic table and data sheet and showed the mole ratios from the equation provided. Weaker responses contained incorrect formulae for nitric acid and unbalanced equations.

Question 27

In better responses, candidates identified Zn as the anode and HgO as the species undergoing reduction or as the cathode. Weaker responses provided only copies of the equations provided in the question.

Question 28

In better responses, candidates clearly demonstrated an understanding of the relationship between stoichiometry and the final concentration calculation. Better responses often included a fully labelled diagram. In the mid-range responses, candidates gave clear and concise descriptions for the titration, including naming the indicator, but some gave limited descriptions for the preparation of the primary standard, not including the use of anhydrous sodium carbonate. Some weaker responses did not include a correct formula for sodium carbonate or a correct formula mass using the periodic table provided. Many weaker responses did not provide a correctly balanced equation for the reaction.

Question 29

- a. In better responses, candidates addressed all aspects of the question, suggesting multiple assumptions and linking them individually to the validity of the process. Assumptions were clearly stated and linked to the amount of sulfate in the fertiliser or the mass of the barium sulfate precipitate.
- b. Better responses included a correct calculation of the mass of the precipitate, stated units and all relevant working was shown in a logical manner. They included a balanced net ionic equation, with states, for the precipitate reaction. In weaker responses, candidates did not perform the multiple steps necessary in the calculation or rounded values before the final answer. In such cases, balanced equations were attempted but confused sulfate with sulfite or sulfide.

Question 30

- a. In stronger responses, candidates included the use of a table to present information to make a comparison. Better responses also clearly identified both addition and condensation polymerisation, described the main features of each and used structural formulae in their equations. The polymerisation of glucose equation required candidates to demonstrate an understanding of the linkage between monomers and better responses showed clearly the elimination of water.
- b. In better responses, candidates clearly identified three polymers and their uses, at least one from ethylene and one from glucose and a link was demonstrated between structure and properties and uses for each polymer. Some candidates explained the structure with the

aid of a diagram. Many mid-range responses demonstrated a good knowledge of the structure, properties and uses of HDPE and LDPE but could not relate the uses of cellulose to its structure and/or properties.

Better responses described the structure of polymers at a molecular level using technical language to explain the relationship between structure and properties. Better responses were specific about the uses, for example plastic bags and cling film rather than using a generic term such as packaging and also specified properties, for example, rigid, hard or tough rather than just strong.

Some weaker responses described polystyrene and/or PVC, but did not clearly distinguish between the properties, structure and uses of polystyrene and expanded polystyrene.

Question 31

- a.
 - i. In better responses, candidates correctly identified and determined the mass of the dissolved solid and calculated the percentage of total dissolved solids in the 250 mL sample to two significant figures. In mid-range responses, candidates correctly calculated the percentage of total dissolved solids but did not provide an answer to two significant figures. In weaker responses, candidates did not correctly identify the total dissolved solid or correctly calculate percentage.
 - ii. In better responses, candidates correctly identified the reagent required as silver nitrate or a solution contain silver (Ag^+) or lead (Pb^{2+}) ions and stated that a white precipitate would be produced with the reaction of chloride ions. Such responses included a correctly balanced equation using correct ionic charges. In mid-range responses, candidates correctly identified silver or lead ion as the reagent required to precipitate chloride ions, but did not correctly balance a chemical equation or did not correctly write the charges on the relevant ions. Weaker responses named incorrect reagents.
- b. In better responses, candidates named a correct ion that causes pollution in waterways, identified its source and included an effect of the identified ion on water quality. Mid-range responses often named a correct ion and stated its effect on water quality. Weaker responses could identify an ion but could not provide an accurate source and provided very general effects or effects not related to water quality.

Section II – Options

Question 32 – Industrial Chemistry

- a.
 - i. In better responses, candidates identified the diagram as the mercury cell and used the diagram and their knowledge of the cell to outline features or processes leading to the production of sodium hydroxide while addressing the extraction of sodium hydroxide.
- b. In better responses, candidates could describe a feature of either cell and compared the electrolysis of molten with aqueous sodium hydroxide, as well as correctly provided half and full equations for both processes. They included clear and explicit comparisons of similarities and differences of molten versus aqueous electrolysis, with correct half and full equations for both processes. These responses had the correct states in all the equations. Weaker responses unnecessarily attempted to provide the chemistry of dilute as well as concentrated sodium chloride electrolysis in their comparison. These responses often strayed from the subject by comparing dilute with concentrated rather than molten with aqueous.
- c.
 - i. In better responses, candidates provided the correct balanced chemical equation, using reversible arrows and also provided the equilibrium constant expression. The working in these responses clearly displayed the use of the 10 L vessel in the calculations.
 - ii. In better responses, candidates correctly interpreted the graph as showing a change to the 'sealed vessel' to be caused only by a temperature change. They also related the temperature change to a new value of K and verified this in terms of change in concentration.
- d.
 - ii. In better responses, candidates indicated a detailed knowledge of the requirements and safety consideration when undertaking this procedure in the laboratory. In weaker responses, candidates often suggested only some safety aspects.
- e. In better responses, candidates included appropriate chemical equations to illustrate the importance of using limestone as well as relating its use to impacts on the environment. These responses also included a clear judgement and were detailed in nature. Weaker responses referred more generally to limestone's importance and impacts and/or did not include a judgement.

Question 33 – Shipwrecks, Corrosion and Conservation

- a. The better responses identified different materials making up the artefact, and related this to the effect of the marine environment.
- b.
 - i. Better responses included a correctly labelled diagram and showed the reduction of water or chloride ions and the oxidation of water half reactions, and combined them for the overall reaction.
 - ii. In better responses, candidates identified $\text{H}_2(\text{g})$ as given off at the cathode via a 'pop' test, or that the cathode was connected to the negative terminal of the power source.
- c. In better responses, candidates related the table to specific properties and uses for all four metals. Better responses identified 'Steel 4' as stainless steel and explained its properties and uses.
In mid-range responses, candidates used the information in the table to predict trends in properties associated with increasing %C and/or the presence of Cr/Ni.
- d.
 - i. In better responses, candidates identified factors that affected the rate of corrosion with correct methods to explain how results

would be observed. They identified the use of a control and demonstrated the effective incorporation of variables to ensure validity of the experiment. In weaker responses, candidates often misinterpreted the question, confused factors that affect the rate of corrosion with those that cause corrosion, or chose non-environmental factors.

- ii. In better responses, candidates identified a method of protection in a marine environment relative to an identified environmental factor. In weaker responses, candidates did not choose a factor identified in d (i) and instead discussed factors present in a marine environment.
- e. In better responses, candidates clearly distinguished between conservation and restoration in relation to copper and wood and demonstrated this by correctly incorporating several correct equations in their explanation. They also clearly compared and evaluated the two materials in a coherent and logical way. Weaker responses produced wrong, irrelevant or no equations and did not adequately address conservation and restoration.

Question 34 – Biochemistry of Movement

- a. In better responses, candidates identified the molecule as adenosine diphosphate and included a balanced equation showing the relationship between ADP and ATP. They also stated that ATP stores chemical energy and releases energy when the molecule is hydrolysed when forming ADP. In mid-range responses, candidates identified the molecule as adenosine diphosphate and related ADP to ATP. Weaker responses identified ADP as ATP.
- b.
 - i. In better responses, candidates identified that the combustion of glucose was a one-step process whereas cellular respiration of glucose contained many steps which release small amounts of energy at every step. In mid-range responses, candidates identified cellular respiration as a multi-stepped process. Weaker responses stated that the cell respiration only slowed down energy production and did not describe cell respiration as a multi-step process.
 - ii. In better responses, candidates identified that the formation of lactic acid causes a reduction in pH. They wrote a correctly balanced equation showing the formation of lactic acid or the dissociation of lactic acid into lactate and hydrogen ions. In mid-range responses, candidates identified that the formation of lactic acid caused a reduction of pH but did not write a balanced chemical equation. Weaker responses did not identify that lactic acid caused a change in pH.
- c. In better responses, candidates stated that amino acid side chains determine the secondary structure of protein. They also included a description of hydrogen bonds, disulfide bonds, electrostatic bonds and dispersion forces and illustrated where the forces occur between the relevant atoms using the diagrams provided. Mid-range responses outlined three of the intermolecular forces involved in the secondary structure of proteins. Weaker responses described polypeptide bonds and the primary structure of proteins.
- d. In better responses, candidates drew an aerobic respiration flowchart containing the glycolysis, TCA cycle and the oxidative phosphorylation stages. They also identified the correct amount of ATP production of each of the three stages listed and the intermediates pyruvate and acetyl-CoA. In mid-range responses, candidates drew a flowchart containing the three stages of aerobic respiration and identified the total output of 38 ATP molecules. In weaker responses, candidates could only draw a simplified flowchart containing two correct stages of aerobic respiration.
- e. In better responses, candidates described and differentiated between type 1 and type 2 muscles and identified the respiration process occurring in each muscle type. They stated and justified specific training programs for both muscle types. They also described the mechanism of muscle contraction. Mid-range responses described the features and differences between type 1 and type 2 muscles and stated examples of training regimes targeting each muscle type. Weaker responses provided features of type 1 and 2 muscles.

Question 35 – Chemistry of Art

- a. In better responses, candidates successfully explained the trend in electronegativity across a period in terms of increasing nuclear charge and movement towards a stable valence configuration. The weaker responses did not identify chlorine due to errors in distinguishing between periods and groups, or not identifying period 3 correctly.
- b.
 - i. In better responses, candidates identified the oxalate ion as a chelated or bidentate ligand.
 - ii. In stronger responses, candidates stated the correct electron configurations of both ions and related the incomplete 'd' subshell configuration to the colour of Cu^{2+} and the complete 'd' subshell in Cu^{+} to its lack of colour. Many candidates did not write correct configuration of these ions or explain the colour/lack of colour using correct colour theory.
- c.
 - i. In better responses, candidates sketched the reflectance spectrum of the pigment as the complement of its absorption spectrum. In weaker responses, candidates confused the colour of pigment, often by not knowing the relationship between wavelengths and colours.
 - ii. Better responses included an outline of the method involved with IR analysis, including an example of a type of pigment this radiation could identify, and a use of this type of analysis.

Weaker responses did not give an accurate outline of the use of infrared radiation in pigment analysis and there was evidence of confusion between absorption and reflectance techniques as well as between UV and IR spectroscopy.
- d. In better responses, candidates included a description of the Bohr model as well as a description of emission spectra, and demonstrated thorough knowledge of how the model explained and predicted hydrogen's spectra.

- e. In better responses, candidates demonstrated thorough knowledge of the names and chemical compositions of a number of pigments used by an identified culture. They also successfully related specific uses of named pigments to their health effects, demonstrating an understanding of the significance of modes of absorption, frequency of use and accumulation within the body.

Question 36 – Forensic Chemistry

- a. In better responses, candidates identified the structures in the diagrams providing an answer which included both a difference between the structures and their origins. In weaker responses, candidates mistook structure A as cellulose or did not provide a difference between A and B based on their chemical structure. Weaker responses often did not distinguish between structural differences and chemical properties.
- b. i. In better responses, candidates understood the principles behind electrophoresis of amino acids and identified that amino acids move not only due to charge and size in an electric field, but also linked movement due to the pH of the buffer solution. In weaker responses, candidates did not recognise that the process of separation is dependent on the charge of the amino acid relative to that of the electric field and the pH.
- ii. In better responses, candidates identified all the amino acids and justified their answers by relating the different movement to not only the different pH of the buffers but also to the correct electrodes. In weaker responses, candidates identified the amino acids without relating their answers to the principles behind electrophoresis.
- c. ii. In better responses, candidates displayed a deep understanding not only of the principles behind emission spectra but also explained their production and linked it to its use in forensic chemistry. In weaker responses, candidates linked the use of emission spectra to identifying elements however they did not explain their production.
- d. i. In better responses, candidates identified four soil properties that a forensic chemist would investigate. Weaker responses indicated a poor understanding of soil properties that could be tested from a crime scene.
- ii. In better responses, candidates clearly identified an inorganic and organic test, providing an appropriate description for its use. Weaker responses did not identify organic properties of soils and provided inappropriate laboratory tests for organic compounds.
- e. In better responses, candidates followed a logical argument and were succinct in their discussion of DNA forensic analysis, giving detailed descriptions of the steps in such an analysis. Such responses included the polymerase chain reaction and electrophoresis which indicated a knowledge and understanding of the underlying principles behind the use of DNA. They linked DNA analysis to its use in forensics providing a description of a range of uses. In weaker responses, candidates provided uses of DNA analysis, however, they did not distinguish between DNA electrophoresis and amino acid electrophoresis as amino acids were confused as being part of DNA. Some candidates were unaware of the wider applications of DNA technology in civil cases, for example paternity in suites and compensation claims.

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